



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.AT.8

Equations and Inequalities Problem Solving

Lesson 7-7 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can model and solve real-world problems using equations and inequalities.

Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Model

Equation

Inequality

Reasonableness

Variable

Constraint



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

- 1 A drawing, equation, or diagram that represents a situation is a

- 2 A math sentence with an equal sign is an .

- 3 A math sentence that compares values with $<$, $>$, $\leq$, or $\geq$ is an

- 4 Checking whether an answer makes sense for the problem is checking its

- 5 A letter that stands for the unknown amount in a problem is a

6

A limit that tells which values are allowed in a problem is a

Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts d ?

A. $d - 7 < 5$

B. $d + 7 < 5$

C. $d - 7 > 5$

D. $d - 7 = 5$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is skipping the key idea: "'Equals' or 'exactly' points to an equation ($=$); 'at least, at most, more than, fewer than' points to an inequality." — always check your work against this rule before you submit.*
8. **Try It 1:** A. $x \leq 20$ — 'At most 20' means 20 or fewer, so $x \leq 20$.
9. **Try It 2:** A. $w = 12$ — Divide both sides by 5: $w = 60 \div 5 = 12$ pounds. Check: $5 \times 12 = 60$ ✓
10. **Exit Ticket:** A. $d - 7 < 5$ — Starting with d donuts and giving away 7 leaves fewer than 5: $d - 7 < 5$.
Solve: $d < 12$. ✓